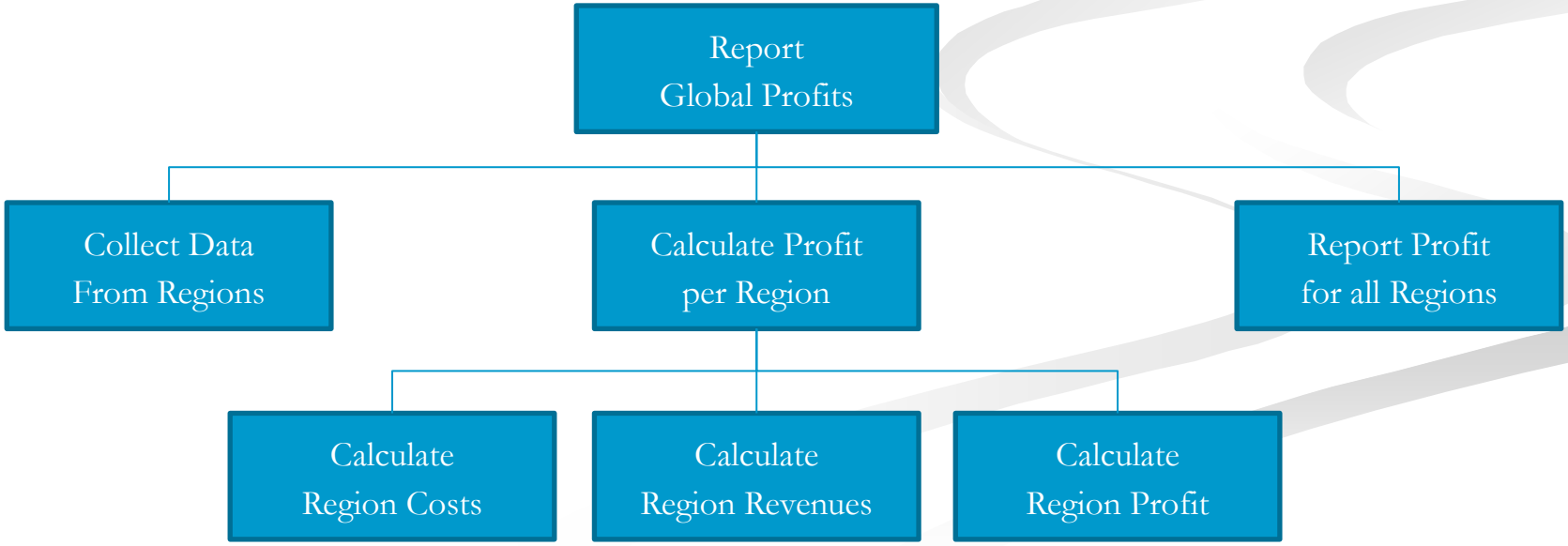
Structured Problem Solving (SPS)

Process Modeling:

Essential Model

Structure of Models Used in Problem Solving
Hierarchy and modularity

- **Hierarchy** decompose tasks / processes and model relationships
  - "Child" activities "belong" to only one "parent" activity
- Decomposition of tasks into **modules** representing **"chunks"** of work
- In programming these modules can be made into functions



Process Modeling in Structured Problem Solving
Hierarchy Chart Guidelines

- Symbols are **rectangles** and **connecting lines**.
- **Rectangles**: a module and the function of that module

- The one rectangle at the top level is the overall problem/model name.
- **Vertical lines** under modules show a **decomposition** of that module

- **Horizontal lines** connects a group of modules in the decomposition of some higher-level module. This higher-level module includes a vertical line below it to connect to the horizontal line of its decomposition.

Process Modeling in Structured Problem Solving
Hierarchy Chart Guidelines (cont.)

- The decomposition of a module must consist of at least two modules (otherwise it would not truly "decompose").
- **Control module:** a module which is decomposed
- **Primitive module:** a module which is not decomposed
- **Sequence:** denoted by the order (from left to right) of the modules on any level.
- **Iteration and selection** have no notation in the hierarchy chart
- Repetition and selection are concerned with how to invoke a module rather than what modules are necessary.

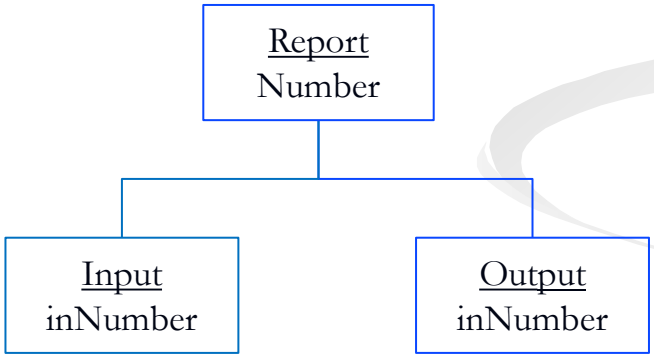
Process Modeling in Structured Problem Solving
Hierarchy Chart module naming / actions

- **Report** - top level module, typically gives a name to reflect the problem being solved
- **Initialize** - a module which stands for the initialization of variables as seen in the walk-through
- **Process** - implies iteration, so the modules on its lower level are considered in the loop
- **Input** - input of the listed variables
- **Output** – output of the listed variables
- **Update** (**Calculate**) - implies a value will be assigned to the listed variables via some formula and/or assignment

Essential Model

Input and output a number

Problem Statement:	I/O Design:
"Develop a program that allows a user to enter a number, and that number is immediately displayed after it is entered.	Output: ▪ inNumber, (number entered) Input: ▪ inNumber, (number entered)



Essential Model

Input single number and output a multiple

Problem Statement:

"Develop a program that allows a user to enter a multiplicative factor and a number. The number multiplied by the factor is displayed immediately after the number is entered.

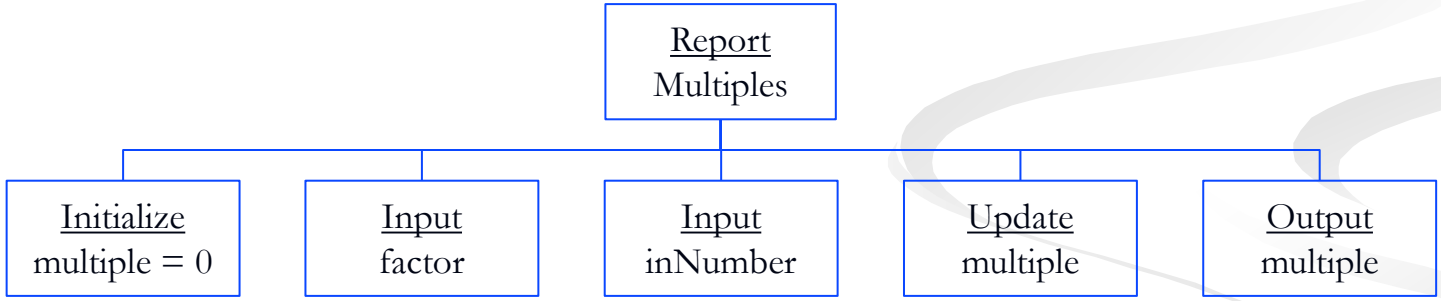
I/O Design:

Output:

- multiple, (number multiplied by factor)

Input:

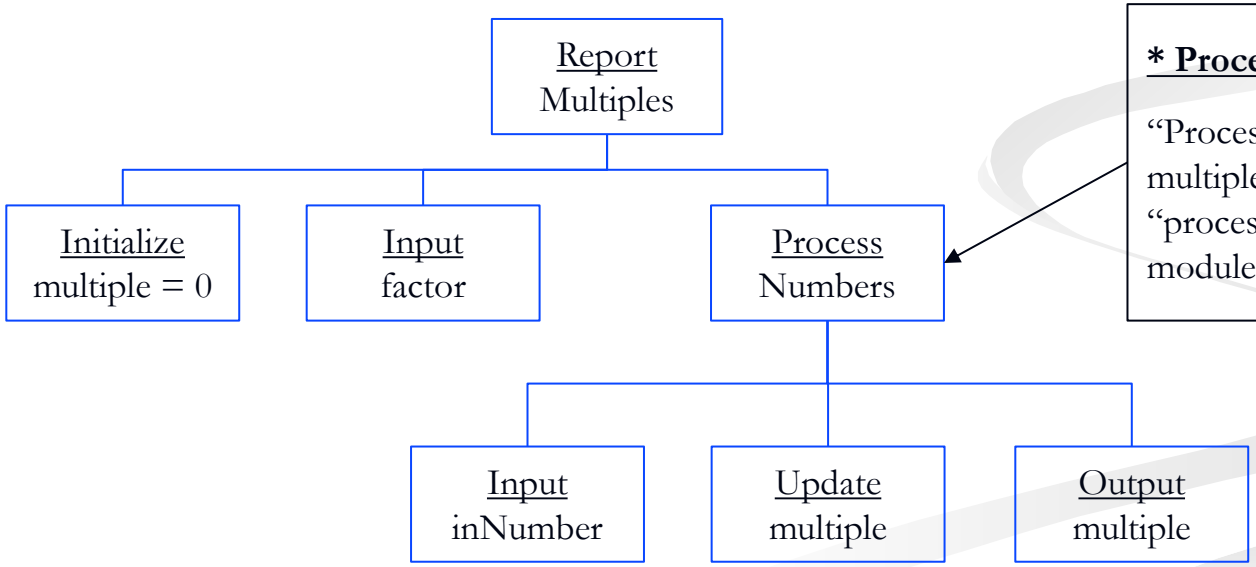
- factor, (number entered as a multiplier)
- inNumber, (number entered)



Essential Model

Input numbers and output multiples

Problem Statement:	I/O Design:
"Develop a program that allows a user to enter a multiplicative factor, and then to enter numbers, one at a time. Each number multiplied by the factor is displayed immediately after the number is entered."	Output: ▪ multiple, (number multiplied by factor) Input: ▪ factor, (number entered as a multiplier) ▪ inNumber, (number entered)



*** Process Modules indicate iteration**.

"Process Numbers" suggests there will be multiple numbers where each one will be "processed" by the decomposition modules of Input, Update, Output.

Essential Model

Cardinality, multiples, and sums

Problem Statement:

"Develop a program that allows a user to enter a multiplicative factor and a count of numbers that will be entered. The user then enters numbers, one at a time until the count is reached. The number multiplied by the factor is displayed immediately after the number is entered. After all numbers have been entered, the program displays separately the sum of all the numbers and the sum of all the multiples."

I/O Design:

Output:

- multiple, (number multiplied by factor)
- sumNumbers, (sum of all numbers entered)
- sumMultiples, (sum of all multiples)

Input:

- factor, (number entered as a multiplier)
- card, (count of numbers to be entered)
- inNumber, (number entered)

